

Locations in a city are vertically differentiated by quality θ . θ can be determined by multiple characteristics, but there is agreement on the ranking. At a location with quality θ , price per square foot of a home in an Nplex (a collection of N homes), each of which has square footage $s(N)$ is $p(N, \theta)s(N)$. We assume that builders are always at a corner solution of building up to $s(N)$ square feet per unit. The function $p(N, \theta)$ includes the relationship between size and number of units implicitly in N . The cost per square foot of an Nplex is $c(N)$, with $c' \geq 0$ because of relatively fixed plumbing and utility costs and fire separation.

The profit from building an Nplex at location θ is thus:

$$\pi(N, \theta) = [p(N, \theta) - c(N)] s(N) \quad (1)$$

Defining $\Delta(n, \theta)$ as the difference in profit from developing $n + 1$ instead of n units, $\Delta(n, \theta) \equiv \pi(n + 1, \theta) - \pi(n, \theta)$, we can identify the effect of location on that gain in profits as an effect on dividing the same amount of space into more units and the effect on location on the benefit to adding more square footage:

$$\begin{aligned} \frac{\partial \Delta(n, \theta)}{\partial \theta} &= \underbrace{\frac{\partial [p(n + 1, \theta) - p(n, \theta)]}{\partial \theta}}_{\text{division effect}} s(n) \\ &\quad + \underbrace{\frac{\partial p(n + 1, \theta)}{\partial \theta} [s(n + 1) - s(n)]}_{\text{addition effect}}. \end{aligned} \quad (2)$$

$$\frac{\partial \pi(N, \theta)}{\partial N} = [p(N, \theta) - c(N)] s'(N) + \left[\frac{\partial p(N, \theta)}{\partial N} - c'(N) \right] s(N). \quad (3)$$

Differentiating with respect to θ :

$$\frac{\partial^2 \pi(N, \theta)}{\partial N \partial \theta} = \underbrace{\frac{\partial^2 p(N, \theta)}{\partial N \partial \theta} s(N)}_{\text{division effect}} + \underbrace{\frac{\partial p(N, \theta)}{\partial \theta} s'(N)}_{\text{addition effect}} \quad (4)$$

The sign of the first, the “division effect” term depends on income sorting at home size N . If higher income households live in better locations at N units per lot, then we would expect $\frac{\partial^2 \pi}{\partial N \partial \theta} < 0$, as sharing space and smaller

spaces are inferior goods. The second term, the “addition effect” is positive if $s'(N) > 0$, that is, if choosing a larger number of units allows more square feet to be built.

1 The division effect in pre-reform Vancouver

Prior to the 2018 Vancouver infill reform, duplex sales were concentrated . . .